Lab Section #9: Hypothesis Testing

Student Lunch Meeting
Nov 2, Monday, 12:15-1:15 pm

What: Part of the Sociology Department's Self-Study. Talking to external reviewers without faculty presence while having pizza and drinks!

When: Nov 2, Monday, 12:15 – 1:15 pm

Where: Powdermaker PH333

Who: For Sociology Majors only.

Extra credit: 0.5 pts (to claim the credit, please sign the attendance sheet and indicate you are from Data 205 on the day of the event.)

RSVP by Friday, Oct 30, 5:00 pm.

https://forms.gle/9CawU7RaKWyXReWe6

Today's Content

Homework 6 Review

Review Concepts from Lecture

Using Posit Cloud for Hypothesis Testing

Discussion for Homework 7

Review: Confidence Intervals

What is a confidence interval?

Confidence interval is an *interval* estimate of a population parameter. It gives a range around a sample statistic.

When do we use confidence Intervals?

We use confidence intervals when we want to estimate a population value using data from a sample. For example, we might survey 100 college students to estimate the average number of hours all college students spend studying.

What do you need to calculate before you construct a confidence interval?

We need to calculate the standard error (SE). It measures how much a sample statistic, such as a sample mean, is expected to deviate from the true population mean. We also need the sample statistic and an appropriate critical value based on the confidence level.

Review: Confidence Intervals

How do you construct a confidence interval?

The general formula is:

Confidence Interval = [sample mean or percentage - (Z-score x SE), sample mean or percentage + (Z-score x SE)]

How do you interpret the results?

Suppose we calculated a 95% confidence interval of [$24,000, $25,980] for the average starting salary. Then we can say, *"We are 95% confident that the true average starting salary of US college graduates lies between $24,020 and $25,980."*

Review: Confidence Intervals

1) Based on the output and your knowledge of the GSS survey, can we make statistical inferences about the US adult population using this sample?

→Is this a 1) large and 2) random sample?

2) Based on the output, what is the sample mean of the average hours Americans worked per week? Please calculate the standard error of the sample mean.

Sample standard deviation

$s = 10.88$

Use the formula

$SE = \frac{s}{\sqrt{n}}$

Substitute the values

$SE = \frac{10.88}{\sqrt{1{,}000}}$

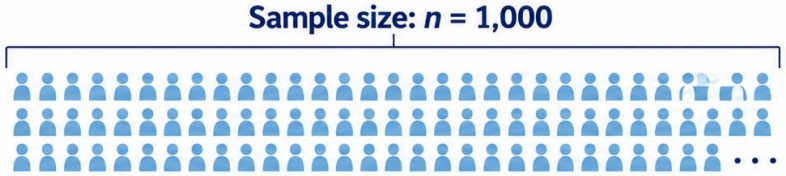
Calculate the square root

$\sqrt{1{,}000} \approx 31.62$

Standard Error

$SE = \frac{10.88}{31.62} \approx 0.34$

1 First take the square root of the sample size, then divide the standard deviation by that result.





Standard error: the avg. distance between all sample means and the population mean



Sample Mean = 42.72

Review: Confidence Intervals

3) Construct the 95% Confidence Interval and interpret the results.

1 Identify the values

Sample mean ($\bar{x}$) = **42.72**
Z critical value (for 95%) = **1.96**
Standard error (SE) = **0.34**

2 Calculate the margin of error

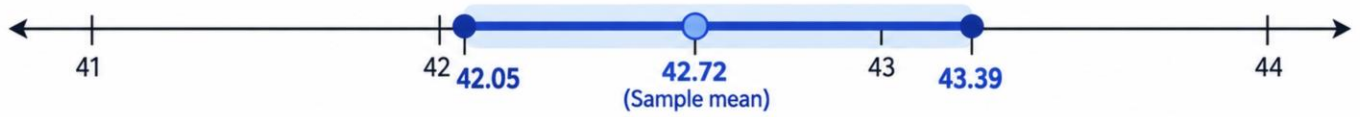
Margin of error (ME) = Z × SE
= 1.96 × 0.34
≈ **0.67**

3 Construct the confidence interval

95% CI = **42.72 ± 0.67**
Lower bound:
42.72 − 0.67 = **42.05**
Upper bound:
42.72 + 0.67 = **43.39**

4 Final answer

95% CI:
(42.05, 43.39)











We are **95%** confident that the true average hours worked per week by American adults lies between **?** and **?**.

Review: Confidence Intervals

4) Based on the output, what is the sample percentage of American adults that have a gun at home? Please calculate the standard error of the sample percentage.

Standard Error of a Sample Percentage: $$SE = \sqrt{\frac{sample\%(100 - sample\%)}{n}}$$

1 Substitute the values

$$SE = \sqrt{\frac{23.10(100 - 23.10)}{1{,}000}}$$

2 Calculate (100 – 23.10)

$$SE = \sqrt{\frac{23.10(76.90)}{1{,}000}}$$

3 Multiply the numerator

$$SE = \sqrt{\frac{1{,}776.39}{1{,}000}}$$

4 Divide by 1,000

$$SE = \sqrt{\frac{1{,}776.39}{1{,}000}}$$

5 Take the square root

$$SE = \sqrt{1.78} \Rightarrow SE = 1.33\%$$

Review: Confidence Intervals

5) Please calculate the 95% CI of the percentage of American adults that have a gun at home in 2006, then **interpret the result**.

95% CI = sample percent ± (Z × SE)

$= 23.10 \pm (1.96 \times 1.33)$

Sample percentage = 23.10%
Standard error (SE) = 1.33 percentage points
Z (for 95% confidence) = 1.96



We are **95%** confident that the true percentage of American adults who have a gun at home lies between ___?___ and ___?___.

Review: Confidence Intervals

6) You could write some R code to replicate the calculation process and cross-check your answers *if you would like*.

```r
#calculating standard error
sqrt((23.10 *(100 - 23.10)))
```

```
[1] 42.14724
```

```r
#constructing confidence interval
23.10 - (1.96*1.33) #Lower Bound
23.10 +(1.96*1.33) #Upper Bound

```

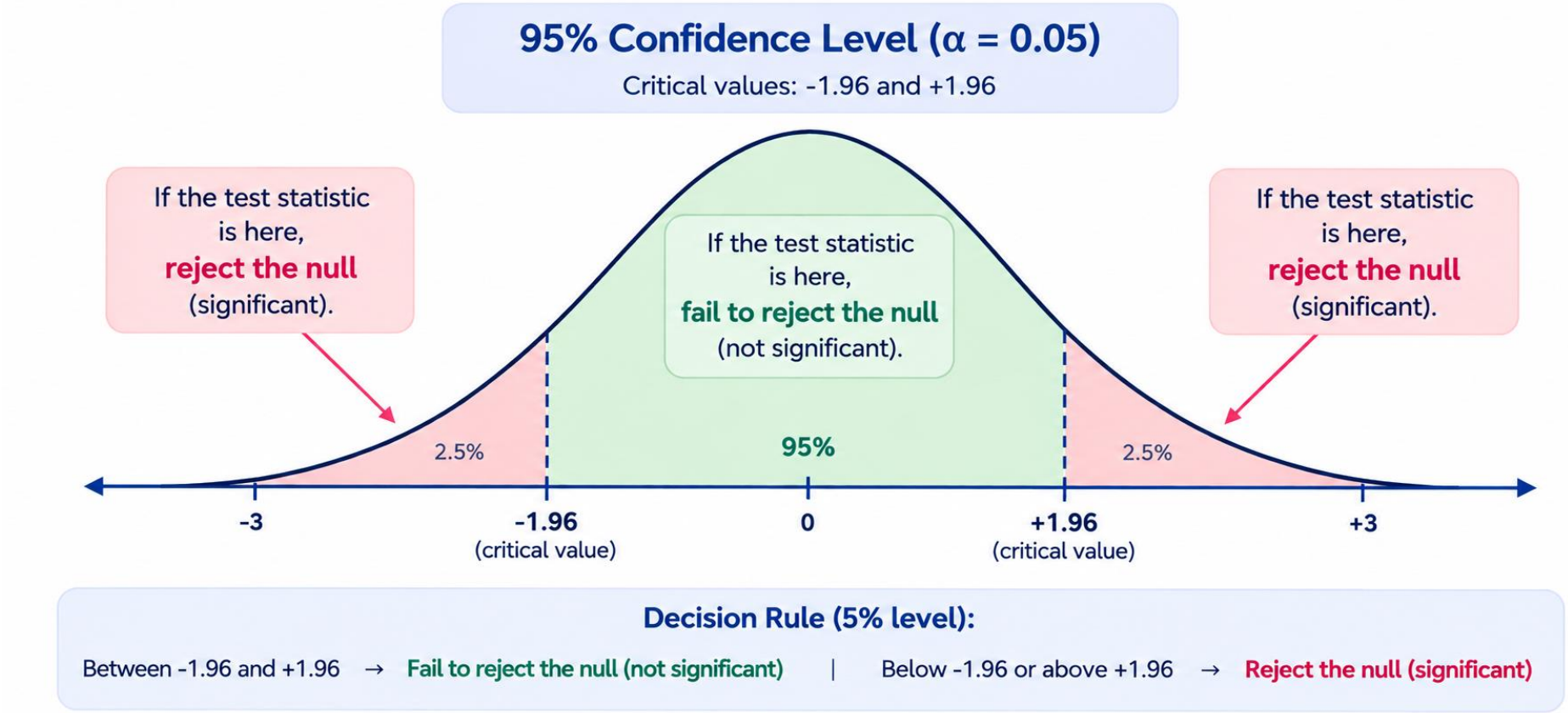
```
[1] 20.4932
[1] 25.7068
```

Review: Hypothesis Testing

Hypothesis Testing: a statistical means of testing a hypothesis about a population, based on sample data.

Hypothesis Testing (also known as significance tests) is used to determine if an observed difference between a sample statistic and a hypothesized population parameter is statistically significant or due to random chance.

Review: Test Statistics



6 Steps of Hypothesis Testing

(0) Organize your information about the hypothesized population parameter (given in the hypothesis), the sample statistic (often found in R output or calculated by yourself), and the confidence level (or the alpha level, given in HW prompts).

(1) Construct Null Hypothesis (H0) and Alternative Hypothesis (H1).

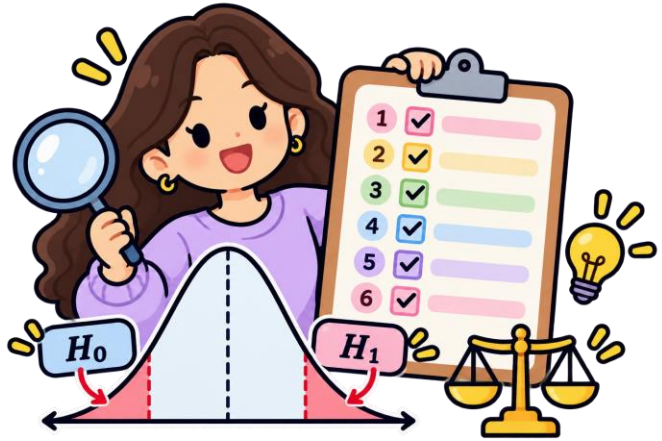
(2) Determine the critical value based on the confidence level (the Z-scores for the 95% or 99% level)

(3) Calculate the test statistic (the specific Z-score of the sample mean/proportion).

(4) Compare test statistic to the critical values: does it fall within or outside of the critical values?

(5) Make decision about whether to reject H0 based on the comparison.

(6) Interpret the results



Construct Hypothesis (Template)

Use this template to write your null and alternative hypothesis. Replace the bracketed information with the details from your specific problem.

H_0: The Null Hypothesis

- H_0 (no difference): $\bar{x} = \mu$. [hypothesized population mean] .
- There is no statistically significant difference between the hypothesized population mean ([hypothesized value]) and the sample mean ([sample mean]) of [variable being studied] .
- Alternatively, you can write it as "$\mu =$ [hypothesized value] ". The true population mean of [variable being studied] is [hypothesized value] .

H_1: The Alternative Hypothesis

- H_1 (significant difference): $\bar{x} \neq \mu$. [hypothesized population mean] .
- There is a statistically significant difference between the hypothesized population mean ([hypothesized value]) and the sample mean ([sample mean]) of [variable being studied] .
- Alternatively, you can write it as "$\mu \neq$ [hypothesized value] ". The true population mean of [variable being studied] is ***NOT*** [hypothesized value] .

Interpretation (Template)

Reject the Null Hypothesis

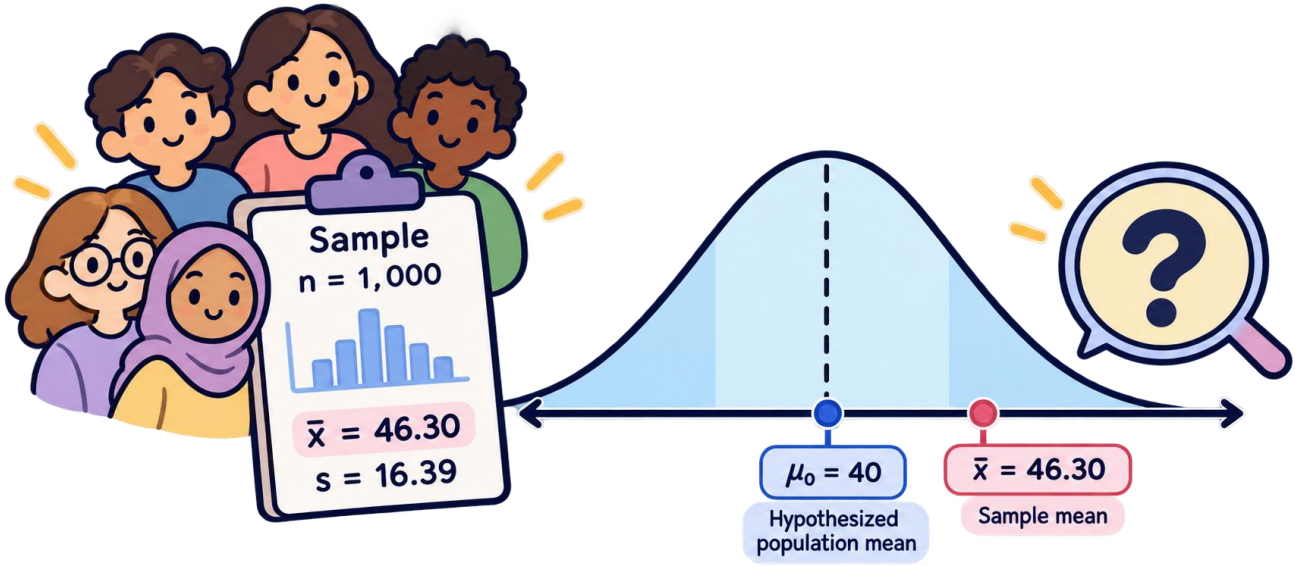
- At the [significance level, e.g., 0.05] significance level (OR: with a [confidence level, e.g., 95%] confidence level), we **reject the null hypothesis** and conclude that there is a **statistically significant difference** between [the variable measured in the sample] ([sample statistic]) and [the hypothesized population parameter or claim] ([hypothesized value]).
- In other words, we are [confidence level, e.g., 95%] confident that the true population parameter for [variable being studied] is ***not*** [hypothesized value].

Fail to Reject the Null Hypothesis

- At the [significance level, e.g., 0.05] significance level (OR: with a [confidence level, e.g., 95%] confidence level), we **fail to reject the null hypothesis.** We do not have sufficient evidence to conclude that there is a statistically significant difference between [the variable measured in the sample] ([sample statistic]) and [the hypothesized population parameter or claim] ([hypothesized value]).
- In other words, we do not have sufficient evidence to conclude that the true population parameter for [variable being studied] is different from [hypothesized value].

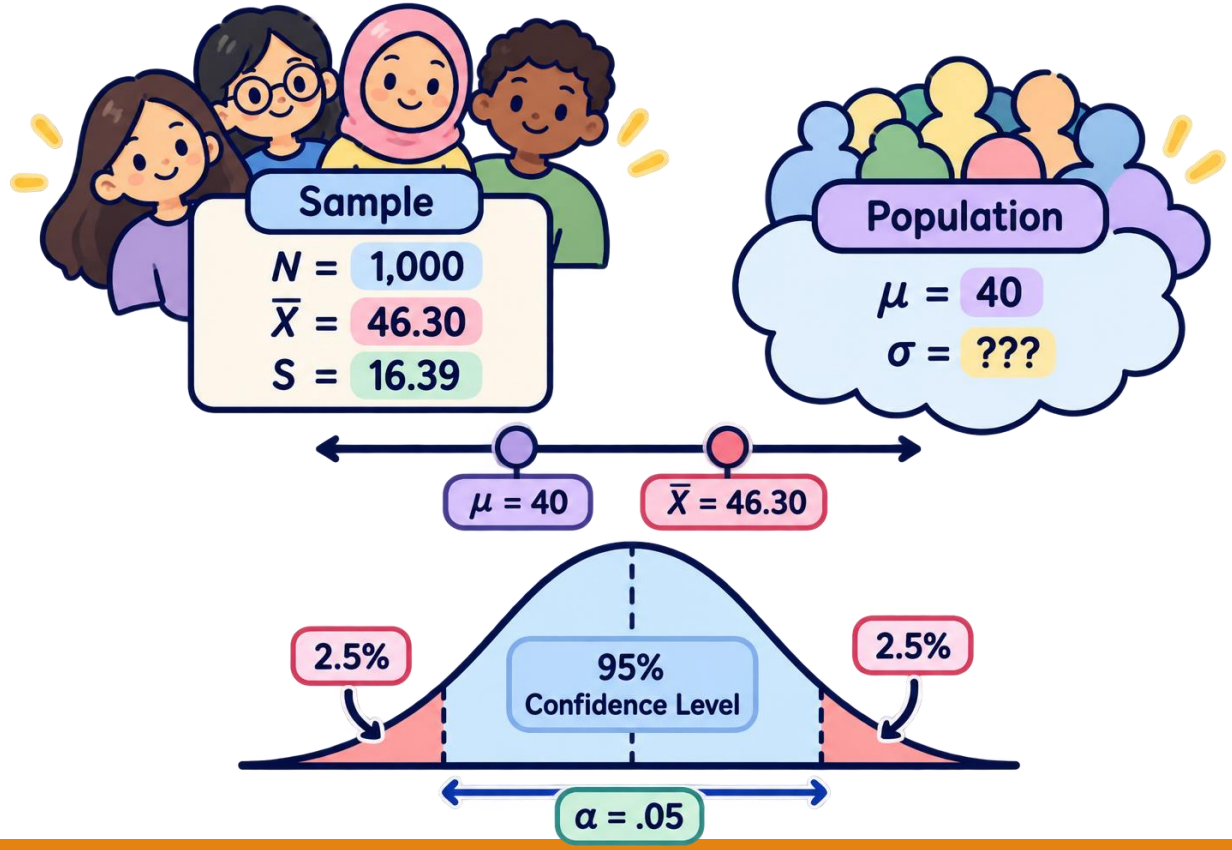
Example: Hypothesis Testing

In a sample of 1,000 respondents, the average age is 46.30 with a standard deviation of 16.39. A researcher hypothesizes that the population mean is 40. Is there a significant difference between the average age of sampled respondents and the researcher's claim about the entire population at the 0.05 significance (alpha) level?



Example: Hypothesis Testing

(0) Organize and label all the information



Example: Hypothesis Testing

Step 1: Construct null hypothesis and alternative hypothesis.

H_0 (Null Hypothesis):

There is no statistically significant difference between the hypothesized population mean age (40 years) and the sample mean age (46.30 years) of the respondents.

H_1 (Alternate Hypothesis):

There is a statistically significant difference between the hypothesized population mean age (40 years) and the sample mean age (46.30 years) of the respondents.



Example: Hypothesis Testing

Step 2: Determine the critical value

What is the confidence level?

Alpha level = 0.05 → Confidence level = 95%

Do we use a Z-Score (1.96 or 2.58) or should we use the T-Table to find the critical value?

N is greater than 100, so we use a Z score.

The critical values of the 95% confidence level:

Z-scores = [−1.96, 1.96]



Example: Hypothesis Testing

Step 3: Calculate the test statistic.

Which test statistic formula should we use? Do you know the population standard deviation?

In this example, the population standard deviation is unknown, so we use the sample standard deviation, s.

$$test\ statistic = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Substitute the values

1. $test\ statistic = \frac{\bar{X} - \mu}{s/\sqrt{n}}$
2. $= \frac{46.30 - 40}{16.39/\sqrt{1{,}000}}$
3. $= \frac{6.30}{16.39/31.62}$
4. $= \frac{6.30}{0.5183}$
5. $test\ statistic \approx \mathbf{12.16}$

Information for this example

Sample (from the data)

$n = 1{,}000$
$\bar{x} = 46.30$
$s = 16.39$

Population (hypothesized)

$\mu = 40$
$\sigma = ?$ (unknown)

Significance Level

$\alpha = 0.05$

Example: Hypothesis Testing

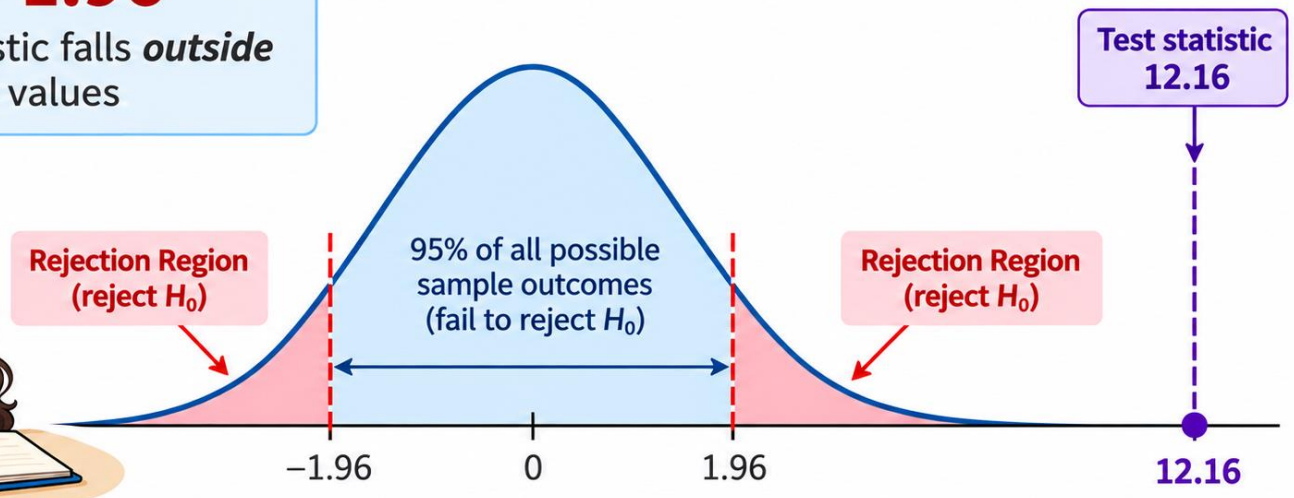
Step 4: Compare the test statistic to the critical value.

- **Critical values:** [−1.96, 1.96]
- **Test statistic:** 12.16

12.16 > 1.96

The test statistic falls ***outside*** of the critical values





Example: Hypothesis Testing

Step 5: Make a decision about the null hypothesis.

If the test statistic falls outside of the critical values → it means that if H_0 were true, it is very unlikely for us to get the observed sample statistic → so we will reject H_0.

In other words, there is a statistically significant difference between the sample mean and the hypothesized population mean.



Apply it to this example:

Test statistic = 12.16 > 1.96

The test statistic falls outside of the critical values, so we reject H_0.

Decision: ***It is very unlikely for us (only 5% chance) to observe a sample mean age of 46.30 years if the true population mean age is really 40 years.***

Example: Hypothesis Testing

Step 6: Interpret Results

At the 0.05 significance level (OR: with a 95% confidence level), we reject the null hypothesis and conclude that there is a statistically significant difference between the average age of sampled respondents (46.30 years) and the researcher's claim about the average age of the entire population (40 years).

In other words, we are 95% confident that the true popular average age is NOT 40 years.



Using R For Hypothesis Testing

You should practice how to do hypothesis testing by hand. But you can also use R to do the test.

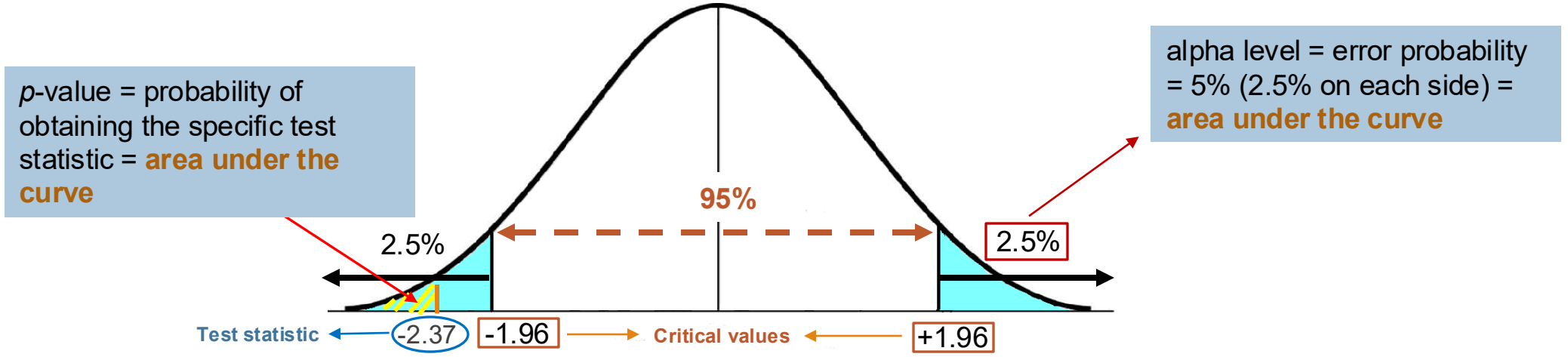
R makes the process easy –

but if you don't understand the step-by-step hand calculation process, you will NOT be able to interpret the R outputs.

P-value is the key to understand R outputs

Two methods of conducting hypothesis testing :

- By hand: Comparing the test statistic to the critical values (e.g., +/-1.96), both of which are Z-scores.
- Use R: comparing the p-value, i.e., probability of obtaining the sample statistic, to the alpha levels (0.05 or 0.01), both of which are probabilities (area under the curve)



The two methods are the same:

- Any sample mean/percentage is associated with a specific test statistic (Z-score, hand-calculated);
- Any sample mean/percentage is associated with a specific *p*-value (probability, R output);
- Decisions about the null hypothesis can be made solely based on the test statistic ***or*** the *p*-value.

P-value is the key to understand R outputs

How to interpret *p*-value?

- ***p*-value < 0.05 (or 0.01)**

  → the probability of getting this specific sample statistic is too small (less than 5% chance) → H_0 is likely *NOT* true

- **reject H_0**

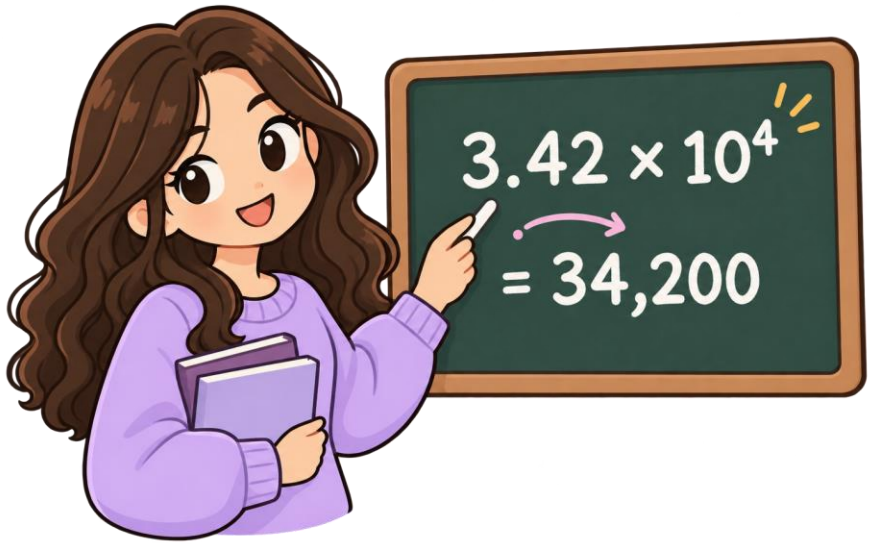
- ***p*-value >= 0.05 (or 0.01)**

  → it's likely to get this sample statistic (>= 5% chance) → H_0 is likely True

- **fail to reject H_0 ("accept" H_0)**

- **You can think of p-value as the probability of H_0 being true (inaccurate, but convenient for understanding.)**

Reading Scientific Notations



Powers of 10 (positive exponents)

10	=	10^1	= e+1
100	=	10^2	= e+2
1000	=	10^3	= e+3

Powers of 10 (negative exponents)

0.1	=	10^{-1}	= e-1
0.01	=	10^{-2}	= e-2
0.001	=	10^{-3}	= e-3

More Examples

- **8334** = 8.334×10^3 = $8.334 \times e+3$ = 8.334e+3
- **6.189 e-4** = 6.189×10^{-4} = 6.189×0.0001 = 0.0006189
- **0.05** = 5×10^{-2} = 5e-2 (so any number times e-3, e-4, e-5, e-6… will all be smaller than 0.05)
- **0.01** = 1×10^{-2} = 1e-2 (so any number times e-3, e-4, e-5, e-6… will all be smaller than 0.01)

Moving the Decimal Point

Positive exponent → move the decimal point **to the right**

7.74521×10^5
= 7.74521
(move 5 places to the right)
= **774 521**

- The exponent is positive (5), so move the decimal point 5 places **to the right**.
- No zeros are needed to fill empty gaps.

Negative exponent → move the decimal point **to the left**

5×10^{-3}
= 0.006
(move 3 places to the left)
= **0.006**

- The exponent is negative (3), so move the decimal point 3 places **to the left**.
- Add in zeros to fill the empty gaps.

Same Example Using Posit Cloud

```
>
> gss <- read_excel('GSS2006.xlsx', sheet='GSS 2006')
>

>
  nrow(gss)
  mean(gss$age)
  sd(gss$age)

  [1] 1000
  [1] 46.295
  [1] 16.38666

> t.test(gss$age, mu = 40

        One Sample t-test

data:  gss$age
t = 12.148, df = 999, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 40
95 percent confidence interval:
 45.27813  47.31187
sample estimates:
mean of x
  46.295
```

H_0: The Null Hypothesis

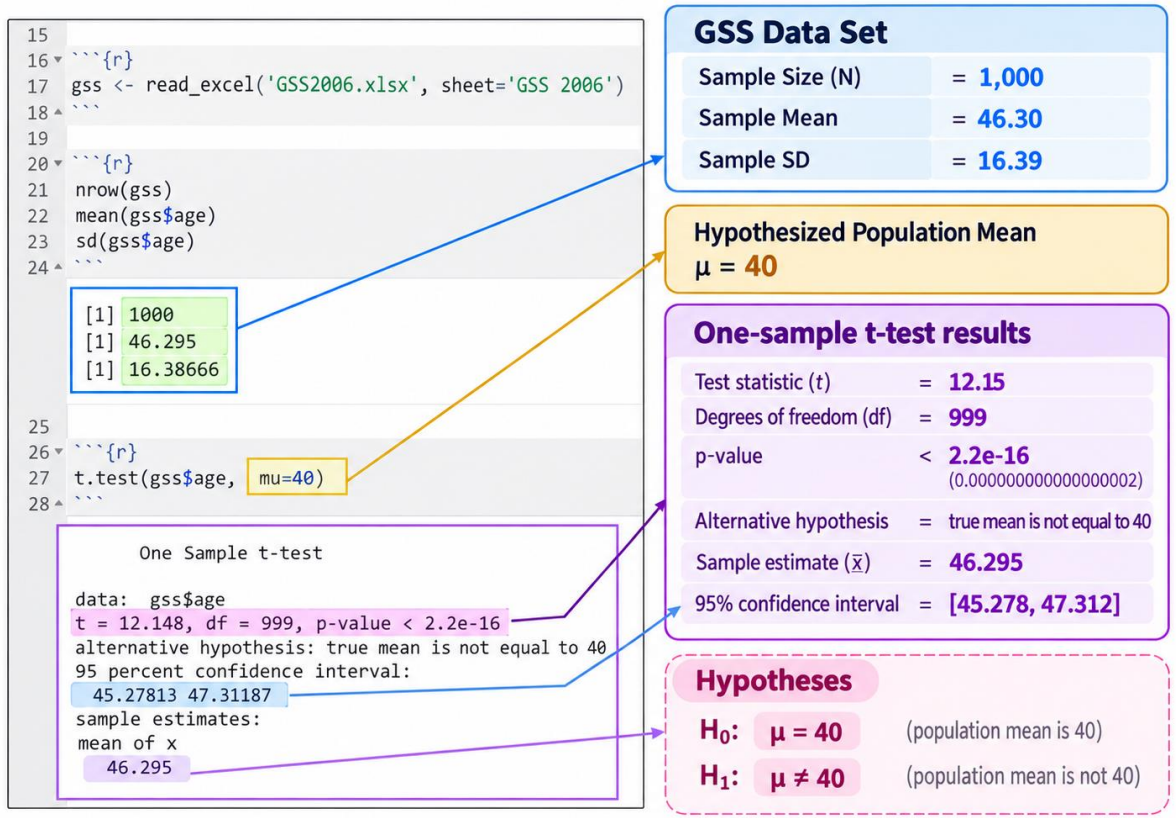
- H_0 (no difference): $\mu = 40$.
- There is no statistically significant difference between the hypothesized population mean (40 years) and the sample mean (46.30 years) of the age of respondents.
- Alternatively, you can write it as "$\mu = 40$". The true population mean of the age of respondents is **40 years**.

H_1: The Alternative Hypothesis

- H_1 (significant difference): $\mu \neq 40$.
- There is a statistically significant difference between the hypothesized population mean (40 years) and the sample mean (46.30 years) of the age of respondents.
- Alternatively, you can write it as "$\mu \neq 40$". The true population mean of the age of respondents is ***NOT*** **40 years**.



Same Example Using Posit Cloud



GSS Data Set

Sample Size (N)	=	1,000
Sample Mean	=	46.30
Sample SD	=	16.39

Hypothesized Population Mean
$\mu = 40$

One-sample t-test results

Test statistic (*t*)	=	12.15
Degrees of freedom (df)	=	999
p-value	<	2.2e-16 (0.00000000000000002)
Alternative hypothesis	=	true mean is not equal to 40
Sample estimate ($\bar{x}$)	=	46.295
95% confidence interval	=	[45.278, 47.312]

Hypotheses

H_0: $\mu = 40$ (population mean is 40)

H_1: $\mu \neq 40$ (population mean is not 40)

By default, R uses a 95% confidence level.

t:	test statistic t-score (z-score is a special case of t-score).
df:	degree of freedom = sample size – 1.
p-value:	the probability of observing the sample mean if H_0 were true.
alternative hypothesis:	H_1 (you need to specify H_0 by yourself.).
95 percent CI:	the CI of the sample mean.
sample estimates:	the sample mean.

p-value: p-value = 0.00000000000000022 **< 0.05**

decision: decision = ***reject the null hypothesis***

Same Example Using Posit Cloud

```
> t.test(gss$age, mu=40)

        One Sample t-test

data:  gss$age
t = 12.148,  df = 999,  p-value < 2.2e-16
alternative hypothesis: true mean is not
equal to 40
95 percent confidence interval:
45.27813  47.31187
sample estimates:
mean of x
46.295
```

By default, R uses a **95%** confidence level.

The sample average age is 46.30 years old. The probability of obtaining this value if the true population average age is 40 is less than **0.0000000000000022**.

In other words, we ***reject the null hypothesis*** and conclude that there is **a statistically significant difference** between the sample average age (46) and the hypothesized population average age (40) at the 0.05 significance (alpha) level.

Discussion for Homework Assignment 7

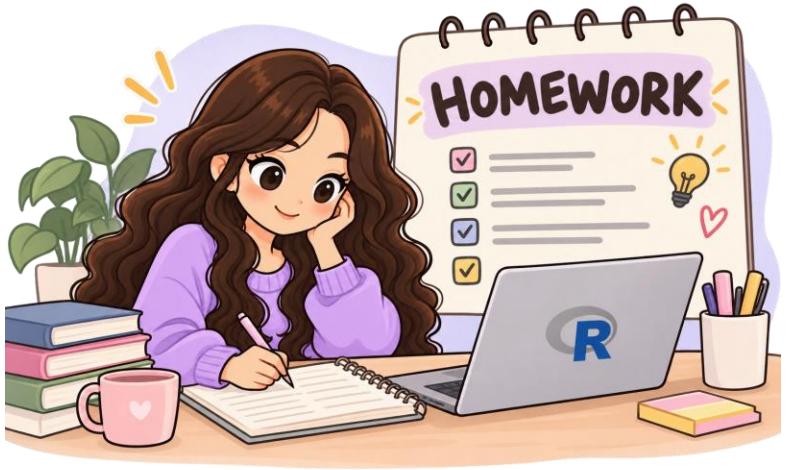
Project Link: https://posit.cloud/content/7102503

Use the link to the Project

Save it as a permanent project in your account

Rename the project, Open the markdown file and resize panes

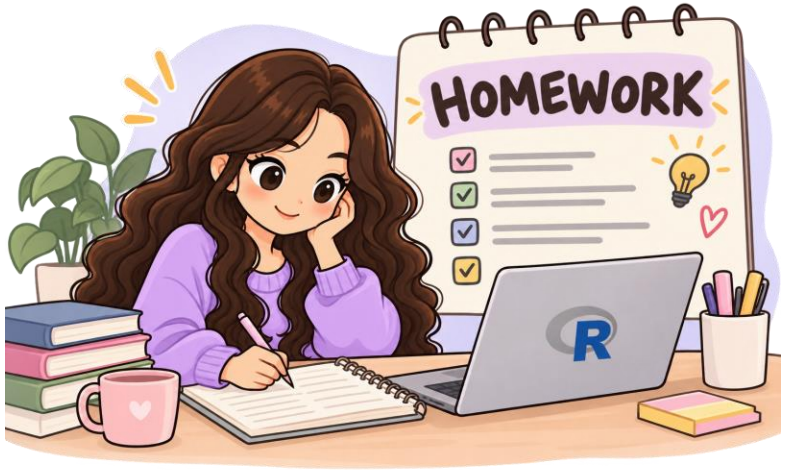
Read all comments and follow the steps in the markdown file:



- First code chunk will load two packages
- Second code chunk will load and inspect the GSS data frame
- Third code chunk will **ask you to write some code** to calculate some statistics for the variable hrs1
- Fourth code chunk will calculate a one sample t-test

Discussion for Homework Assignment 7

Project Link: https://posit.cloud/content/7102503



Change the author (line 3) and date (line 4) of the markdown file

Knit the Markdown file and save it to your computer/device

Open the Assignment in Brightspace

Answer the questions in Brightspace

Submit the knitted file in Brightspace

Due on November 1st at 11:59pm

Q1. Hypothesis Testing by hand (6 steps), hypothesized population mean = 40.

Q2. Hypothesis Testing in Posit Cloud, hypothesized population mean = 43.